

Ans 1. Let $S(n) = n^3 + 2n$ is divisible by 3.

Step I For $n=1$

$$1^3 + 2 \cdot 1 = 3 \text{ which is divisible by 3}$$

$\Rightarrow S(1)$ is true.

Step II

Let $S(k)$ is true

$\Rightarrow k^3 + 2k$ is divisible by 3. holds good.

$$\Rightarrow k^3 + 2k = 3s \text{ for some } s \in \mathbb{N}.$$

Step III We have to show that $S(k+1)$ is true.

Consider $(k+1)^3 + 2(k+1)$

$$= k^3 + 1 + 3k^2 + 3k + 2k + 2$$

$$= (k^3 + 2k) + 3(k^2 + k + 1)$$

$$\textcircled{a} \Rightarrow = 3s + 3m, \text{ where } m \in \mathbb{N}$$

$$= 3(s+m)$$

which is divisible by 3.

By P.M.I. $S(n)$ is true for all $n \in \mathbb{N}$.

Ans 2 Let $S(n) \equiv 8^n - 3^n$ is divisible by 5

Step I. $S(1) = 8 - 3 = 5$ is divisible by 5
 $\Rightarrow S(1)$ is true

Step II Let $S(k)$ is true

$\Rightarrow 8^k - 3^k$ is divisible by 5

$\Rightarrow 8^k - 3^k = 5S$ for some $S \in \mathbb{N}$.

Step III we will show that $S(k+1)$ is true.
 $\Rightarrow$ i.e. $8^{k+1} - 3^{k+1}$ is divisible by 5.

* Now consider

$$\begin{aligned} 8^{k+1} - 3^{k+1} &= 8 \cdot (8^k) - 3 \cdot (3^k) \\ &= 5 \cdot 8^k + 3 \cdot 8^k - 3 \cdot 3^k \\ &= 5 \cdot 8^k + 3[8^k - 3^k] \\ &= 5 \cdot 8^k + 3 \cdot 5S \quad [\text{using (a)}] \\ &= 5[8^k + 3 \cdot S] \end{aligned}$$

which is divisible by 5.

Therefore by P.M.I, $S(n)$ is true for all $n \in \mathbb{N}$.

Ans 3. Step 1. $S(1) \equiv 1^2 = \frac{1(1+1)(2+1)}{6} = 1 \Rightarrow S(1)$ is true

Step II Let $S(k)$ is true

i.e. $1^2 + 2^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$ holds true.

Step III We will show that $S(k+1)$ is true. We will show that

$$1^2 + 2^2 + \dots + (k+1)^2 = \frac{(k+1)(k+2)(2k+3)}{6}$$

consider $1^2 + 2^2 + \dots + k^2 + (k+1)^2$

$$= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 = (k+1) \left[\frac{k(2k+1)}{6} + (k+1) \right]$$

$$= (k+1) \left[\frac{2k^2 + 7k + 6}{6} \right] = \frac{(k+1)(k+2)(2k+3)}{6}$$

$\Rightarrow S(k+1)$ is true

Let $S(n) \div 2^n > n^3$ for $n \geq 10$
Ans 4. Step I

$$S(10) : 2^{10} = 1024 > 10^3$$

Hence $S(10)$ is true.

Step II Let us assume that $S(k)$ is true.

$$\text{i.e. } 2^k > k^3 \text{ for } k \geq 10.$$

Step III We have to show that $S(k+1)$ is true.

$$\text{i.e. } 2^{k+1} > (k+1)^3$$

Consider

$$2^{k+1} = 2^k \cdot 2 > k^3 \cdot 2 \quad (\text{using II Step})$$

$$> (1 \cdot 1)^3 k^3 \quad [\because (1 \cdot 1)^3 < 2]$$

$$> \left[1 + \frac{1}{10}\right]^3 k^3 \quad \text{--- (A)}$$

* Now Consider $k \geq 10$

$$\frac{1}{k} \leq \frac{1}{10}$$

$$1 + \frac{1}{k} \leq 1 + \frac{1}{10}$$

$$\left(1 + \frac{1}{k}\right)^3 \leq \left(1 + \frac{1}{10}\right)^3$$

$$\left(1 + \frac{1}{k}\right)^3 \cdot k^3 \leq \left(1 + \frac{1}{10}\right)^3 k^3 \quad \text{--- (B)}$$

using (B) in (A), we get

$$2^{k+1} > \left(1 + \frac{1}{k}\right)^3 \cdot k^3$$

$$= \frac{(k+1)^3}{k^3} \cdot k^3$$

$$\Rightarrow 2^{k+1} > (k+1)^3 \Rightarrow S(k+1) \text{ is true}$$

Hence $S(n)$ is true for all $n \in \mathbb{N}$.

Ans. 5.

p : n is even

q : n^2 is even

$p \rightarrow q$: if n is even then n^2 is even

Let p is true $\Rightarrow n = 2m$ where $m \in \mathbb{Z}$

$$\begin{aligned}\Rightarrow n^2 &= 4m^2 \\ &= 2 \cdot (2m^2) \\ &= 2k \quad (2m^2 = k)\end{aligned}$$

$\therefore n^2$ is even

$\Rightarrow q$ is true

Hence $p \rightarrow q$ is true

i.e. if n is even then n^2 is even

Ans. 6.

p : x is difference of square of two consecutive integers

q : x is odd

Let p is true $\Rightarrow x = (k+1)^2 - k^2$, where $k \in \mathbb{Z}$

$$= 2k+1$$

$\therefore x$ is odd

$\therefore q$ is true

$\Rightarrow$ difference of square of two consecutive integers is odd.